

PURVA SUSHIL FIRODIA

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SUMMARY

- Machine Learning Engineer with 3+ years of experience delivering AI/ML solutions in finance, SaaS, and enterprise platforms with measurable business impact.
- Expertise in designing, deploying, and scaling **ML models (Classification, NLP, LLMs, Time Series Forecasting, Computer Vision)** using **PyTorch, TensorFlow, and Scikit-learn**, improving accuracy and reducing manual effort by up to 40%.
- Strong background in **data engineering (ETL pipelines, data modeling, Snowflake, Airflow, Apache Spark, Kafka)** and cloud-native deployments (**AWS SageMaker, Bedrock, Lambda, Kubernetes**).
- Proficient in building real-time analytics dashboards (**Power BI, Tableau, Plotly**) and delivering actionable insights for business optimization, leading to 15–25% improvements in efficiency, revenue, or decision-making.
- Hands-on experience with **Salesforce (APEX, Lightning Web Components, API integrations)**, integrating AI-driven analytics into **CRM workflows** to enhance customer engagement and retention.

SKILLS

Languages: Python (NumPy, Pandas, sciPy, PySpark), R (ggplot2), SQL, C++, Java, Bash, Object-Oriented Programming (OOP)
Machine Learning & AI: scikit-learn, TensorFlow, PyTorch, Keras, XGBoost, SHAP, LightGBM, OpenCV, LLMs (Hugging Face, Prompt Engineering), Gen AI, Time Series Forecasting (ARIMA), NLP (spaCy, NLTK, BERT), Computer Vision, Feature Engineering, Deep Learning (CNN, RNN, LSTM, GANs), Classification, Regression, Clustering
Data Engineering: dbt, Airflow, Snowflake, PostgreSQL, MySQL, MongoDB, Data Warehousing, ETL/ELT Pipelines, Data Modeling
Data Visualization: Tableau, Power BI, Excel (PivotTables, VLOOKUP, Power Query), Matplotlib, Seaborn, Plotly
DevOps & Cloud: AWS (S3, EC2, SageMaker, Lambda, Glue, Bedrock, Redshift, Athena, RDS, CloudWatch), Docker, Kubernetes, Jenkins, Apache Spark, Apache Kafka, Git, GitHub Actions, CI/CD, Terraform
Other Skills: Agile/Scrum, Flask, Django, FastAPI, REST APIs, Salesforce (APEX Code, Lightning Web Components, API)

EXPERIENCE

Northern Trust	United States
Machine Learning Engineer	January 2025 – Present
- Designed, trained, and deployed fraud detection models (XGBoost, PyTorch, SQL feature engineering) on 100M+ transactions, reducing false positives by 22% and saving the bank \$1.5M annually. - Developed time series forecasting pipelines (ARIMA, LSTM, Python, Airflow) to predict client cash flow and liquidity needs, improving forecast accuracy by 18%, which enhanced treasury planning decisions. - Built ETL pipelines in Snowflake + Airflow to process 50M+ daily transactions, optimizing SQL queries and cutting pipeline latency by 40%, achieving 99.9% reporting accuracy for compliance. - Automated Python + SQL-based anomaly detection scripts for high-value trades, reducing manual monitoring time by 35% and strengthening fraud risk controls. - Designed Power BI dashboards integrated with SQL backends to visualize portfolio risk, client segmentation, and anomalies, accelerating executive decision-making by 30%. - Integrated Salesforce APIs with ML risk models to push real-time fraud alerts into CRM workflows, enabling relationship managers to take proactive action and improving customer trust scores by 15%. - Collaborated with cross-functional teams (Risk, Treasury, IT) under Agile/Scrum, ensuring on-time delivery of 95% of ML projects while maintaining strict compliance and audit controls.	
Sage SoftTech	India
Data Scientist	January 2021 – July 2023
- Developed NLP-driven classification models (spaCy, BERT, XGBoost, Python, SQL) for 200K+ support tickets, improving query routing accuracy by 28% and reducing SLA breaches by 15%. - Built customer churn prediction models (Random Forest, Gradient Boosting, SQL feature pipelines) with an F1-score of 0.91, enabling targeted retention campaigns that reduced churn by 12%. - Designed real-time data pipelines with Apache Kafka + PySpark to ingest 1M+ daily events across CRM and sales, reducing data lag from 12 hours to <15 minutes, improving reporting speed. - Engineered computer vision models (CNNs, OpenCV, Python) for invoice digitization, anomaly detection, and fraud flagging, cutting manual data entry time by 60% and reducing errors. - Deployed ML models on AWS SageMaker, Lambda, and Kubernetes using CI/CD pipelines (Docker, GitHub Actions), improving scalability and reducing deployment cycle time by 40%. - Partnered with Salesforce engineering team to embed AI-powered recommendations into Lightning Web Components (LWC), driving 10% uplift in upsell conversions for sales teams. - Conducted SQL-based A/B testing and statistical analysis on customer segmentation strategies, generating insights that improved targeted campaign ROI by 17%.	

EDUCATION

University of Colorado Boulder, Boulder, Colorado August 2023 – May 2025
Masters of Science in Data Science
<i>Coursework: Machine Learning, Natural Language Processing (LLM Techniques), Data Mining, Statistical Inference, Deep Learning, Data Center Scale Computing, Information Visualization, Neural Nets & Deep Learning, Predictive Analysis</i>
Savitribai Phule Pune University, India September 2018 – August 2022
Bachelor of Engineering in Computer Engineering
<i>Coursework: Data Structures, Algorithms, Database Systems, Object Oriented Programming, Soft Computing & Optimizing Algorithms, Data Analytics, Data Warehousing, Business Intelligence, Machine Learning & AI</i>

PROJECTS

Next-Gen E-Commerce Platform with Serverless Microservices

- Designed a cloud-native e-commerce platform with a serverless backend using AWS Lambda and API Gateway, handling 500+ concurrent requests efficiently.
- Developed a responsive React.js frontend, deployed on AWS S3, achieving 25% faster page load times via CloudFront, and integrated MongoDB and Cloudinary, improving database performance by 20% and media efficiency by 15%.
- Implemented PayPal payment gateway, streamlining secure transactions, enhancing checkout experience, and improving customer satisfaction.

Customer Behavior Analysis for Business Optimization

- Developed Random Forest model to predict high-value and anomalous customer segments with 94% accuracy, leveraging data mining, predictive analytics, and A/B testing insights.
- Created interactive dashboards in Tableau to visualize customer behavior, enabling data-driven marketing strategies and increasing campaign conversion rates by 20%.
- Automated SQL pipelines reducing data preparation time by 30% & streamlining reporting, showcasing experience with ETL processes.

Speech-Based Depression and Gender Classification Analysis Using Machine Learning

- Developed an XGBoost-based machine learning model to classify depression and gender from speech patterns, achieving 95% accuracy with balanced True Positive Rates (TPR) across genders.
- Applied advanced feature engineering and handled missing data strategically, improving model robustness by 20%.

Traffic Sign Image Recognition and Classification using CNN.

- Designed and trained CNN-based classifiers using Keras and TensorFlow, achieving 90%+ precision on 10K+ multiclass road sign images.
- Applied OpenCV and Python for preprocessing, data augmentation, and post-processing to improve classification speed and robustness.

CERTIFICATIONS

- [AWS Certified AI Practitioner](#)
- [AWS Certified AI Practitioner Early Adopter](#)
- [\[NEW\] Ultimate AWS Certified AI Practitioner AIFC01](#)
- [Expressway to Data Science: Python Programming](#)
- [Python Packages for Data Science](#)
- [Android Basics in Kotlin](#)
- [Salesforce Developer Virtual Internship](#)

PUBLICATIONS

Traffic Sign Image Recognition and Classification using Deep Convolutional Neural Network. | [Link](#)

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